

ActInf Livestream #010.2: "A variational approach to scripts" (2020)

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hello everyone welcome to the active inference live stream this is active inference live stream 10.2 it is December 15 2020 and we are really excited to be having this discussion with all of you today welcome to the active inference lab everyone we are an experiment in online team communication learning and practice related to active inference you can find us at our website on Twitter at email our YouTube channel or our public qbase team and username this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and as far as video etiquette for live streams remember to mute if there's noise in your background raise your hands so that we can hear from everyone on the stack and use respectful speech Behavior just as a quick little announcement SLO of process all the rest of the meetings for the rest of 2020 are at 7:30 to 9:00 a.m. PST last week we had 10.1 on the scripts paper and today we're having a follow-up discussion on the scripts paper 10.2 and then for the final two weeks of December we're going to be reading the paper sophisticated effective inference so check out the dates and um contact us if you want to participate in these final discussions or in 2021 or if you have an idea for an event that we could do on the actm stream we have a lot to announce coming up soon and one good place to stay apprised of that would be on our website where you can find this call for collaboration link so if you go to active inference dorg and check out this aweber link you'll see some information about collaborating in 2021 on the live stream and on other really cool projects great so today in ACM 10.2 we are going to start with introductions and warm-ups and then we're going to go through 10.2 we're going to be talking about the variational approach to scripts paper that we've been talking about for the last week as well and check out 10.0 if you want some context and check out 10.1 for last week's awesome discussion we'll go through the Asim claims abstract and road map today and then there's some new sets of questions about types of scripts and how do scripts look and other questions and then we'll um look through the figures which is something that we brought up last time and

next week there's going to be a different paper so get in touch if you want to participate all right thanks for that now on to the intros and warm-ups so for the introductions and warm-ups let's introduce ourselves by giving a short introduction or check-in and then passing it to somebody who hasn't spoken I'm Daniel I'm a postdoc in California and I will pass it to Blue I am blue Knight I am an independent research consultant based out of New Mexico and I will pass it to RJ my name is RJ cord uh I'm on the east coast of the United States and I'm mostly doing research and organizational Dynamics intelligence analysis and narrative Warfare and I'll pass it to Shannon if Shannon here we'll come back to Shannon come back uh Alex then uh thanks I'm Alex I'm located in Moscow Russia and I'm a researcher in systems management school and I pass it to Sasha hi I'm Sasha I'm uh a graduate student also in California and I study neuroscience and development and I will pass it to Steven hello I'm Steven I'm based in Toronto in Canada I do a lot of work with participatory theater and S of participatory development workshops and I'm doing a practice-based PhD at the moment and I'm gonna pass it to Ivan hello my name is Ivan uh I'm based in Moscow I'm interested in act in for in in case of system thinking and I pass it to Shannon Shannon cool well I think Shannon will come back online whenever she's ready so for now we can just continue on to the warm-up questions for these questions feel free just to raise your hand we're happy to hear from anyone so one question since many of you were here last week is what is something that you thought about differently after our conversation last week and so while people are raising their hand um I just really like this way of thinking about scripts and genres for our encounters and at the life trajectory level and at the sort of sub scene level what genre are we drawing our scripts from RJ uh so after last week's conversation uh what I ended up really thinking about is happens when we lose our scripts so it it seems like scripts are are um a framework for representing like the like how how we how we move through these complex social interactions um so what happens when say like in the Spanish Inquisition where it was a process by which they were actively trying to remove scripts that people had um what happens in the absence of them uh when we tear them away so that that's that's I've been thinking about that a lot the last week interesting question blue I've been thinking really a lot about um like the transparency of active inference versus like other types of like deep learning and how they're kind of a blackbox and really what RJ uh had said last week about how active inference is more like a blanket of black boxes I thought that that was pretty cool and I've been like kind of you know twirling that round in my head for for the last week cool and it's almost like um if the sentence Oh go ahead Sasha um uh kind of the opposite of RJ for the last week I've been thinking uh

what's not a script um and can there be an absence of a script because um we're we're just as apt to follow another set of directions as we are um the cultural set of directions that we get for interactions so um that's been kind of uh vexing because it's it's similar to the problem with um thinking about free energy and active inference that once you see it then you wonder like what's not in that category and um how do we kind of draw the correct boundaries around uh the definitions of these terms because yeah once you see scripts you really see it around around you in every situation yeah to your point there Sasha it's like if a firsttime ever social experience or a very ritualized one if the scripts concept can include both of those as well as the internalist and externalist concept it's kind of like how many grains of sand in a heap like how reproducible or how strong does the pattern of a social encounter have to be before we're going to deem it scripts of a certain kind blue so to kind of to piggyback on what you and Sasha both saying like I've been thinking about like is is maybe everything is a script like is a chair a script really like I mean a chair has like a set role and like a a thing we do with it or it can you know take on a new like script so like what what about all of these like objects in our everyday Society really like once you once you see a script like what is not a script like I I I have a hard time like you know unfolding that well it's like the wood is chairing you know the chair is a noun but it's a it's a process and it was it built and it will eventually not be a chair and so while it's chairing we can draw a blanket you know put a seat cover on it and use it in the functional interface of being a chair and so especially when things become um sufficiently compartmentalized or interface so that they have a functional way of dealing with them instead of just getting lost in the becoming and Unbecoming of chairs we can start to think all right well yes that is happening but also there's this stable pattern that I can sit on that's why we're calling it a chair and so it's like this emergence of reproducible function and pattern out of continual becoming and Unbecoming and so that is to the point with a lot of discussions about the marov blanket or the frisen blanket whatever it ends up being in these questions about ergodicity as well like does everything by virtue of having inputs and outputs is everything going to be under this umbrella or is this kind of umbrella only going to be over things that have um input output relationship ship of some type over some spatial or temporal scale Stephen I think this also made me think about where sense making comes in um and how this approach to scripts it kind of it helps extend the active inference work with deontic cues you know they talk about the traffic lights and a lot of other things in the models and this starts to extend these sort of top down cues um into narratives and stuff like that but there's still the the more existential kind of participatory sense making and the kind of existential sense

making processes which are maybe um they're still open to sort of somehow integrate with this more scripted kind of way of making sense of the world so I think that's that's something that I'm interested in in in where do they start to sort of mesh or override each other in different ways cool all right let's go to the next warm-up question how could a different idea of scripts relate to how we regulate the behavior of ourselves and others so depending on how somebody thinks about scripts how might they be likely to treat their own behavior differently and how might they be likely or willing to engage in different kinds of regulations of other people's behavior so let's just say that there's some prop in a script so something like a uh artifact or an object so just like to RJ's example about wanting to uproot a script so then you could say well you can't use this symbol or you can't use this combination of words or we're not allowing anything from this genre of script and we're going to know it because it's going to have this style and um so depending on how we formalize these scripts maybe in order to regulate other people's deployment of scripts we might take very specific um subcomponents like a certain set of words or an artifact and then say well this is or isn't ruled in RJ uh yeah so so on on this um uh one one thing uh going going back to the last question just you kind of uh uh stole the words out of my mouth what I wanted to say which was is a a uh is a chair a script for wood um but uh I I also want to note like on this is I I I got to have a conversation yesterday with Eric Sapp from public democracy and a lot of the stuff they've been doing uh as I mentioned to you before Dan is like um using like traditionally what what would be used for for advertising uh for for for like public good so uh what they've done in one case was was be able to uh use the same tools that they would use to Target advertising to um uh Target recent people who have very recently picked up a heroin addiction um looking at their their behaviors online be able to figure out that that they're that they've recently picked up this addiction or or that they very likely had and then give them on-ramps toward treatment at the time when they are most likely um uh able to to move toward accepting those types of treatments they're they're not they're not very deep into the addiction yet so um I I brought up scripts to him uh and we we talked a little bit about it and it seems like their look at it is like yeah it's it's almost like one if if you can give people the idea um that um they're using these scripts and that they can change them and that they're creating them in real time uh or or adapting to them in real time then um you kind of have the self-awareness that can kind of help you um moving down like to we we talked about radicalization last week like if you're aware of of of the scripts that are in the space may maybe you can you can be self-aware when you're using them uh so I think that's one thing and then um in addition like recognizing when you're about to hit a threshold where

you're about to like start using a script rather than just some Behavior which kind of ties to what Sasha was saying is like at what point do we call something a script so I'm not sure but but anyway yep um it it reminds me of the Gateway idea like gateway drug or Gateway website or Gateway meme whatever it is it's like using that metaphor it's like whoa if you go through the Gateway if you pass through you're not coming back it's all downhill from there well that's one script it's a narrative for how you could proceed through different types of experiences but could there be a sort of like visitation metaphor it's like hey you're going to visit this landia and then your family is still going to be waiting for you and they all want you to become uh safe and come home for example and then that could be a attractor that takes some of the same setup it's a similar genre at least in the beginning of the movie but then it goes a different way and so that definitely is a is a nice point there okay and then our last uh warm-up question with something that you're still wondering about or something that you'd like to have resolved by the end of today's discussion I really like this question also about the what's not a script or defining by defining what isn't another thing that wrote I wrote down to ask about is the the transparency of the concept is something that we'll probably return to many times and is it really more transparent to subc compartmentalize it into a lot of smaller black boxes or are we doing something quite distinct from that what else does anyone think like what could be a fun area to take this script's concept in or what could be somewhere we could go in the discussion but if no thoughts here then no worries because there's a bunch of questions that will be asked in the later stages so good okay Shannon I hope that you can hear us or rejoin when you can but let's get into it and then anyone just raise your hand when things come to you and we'll go from there cool so the paper that we are discussing this week as with last week is a variational approach to Scripts and we had Mao last week on the panel which was great the aims and claims of the paper we went through those in 10.0 and heard it from the authors themselves last week but the big question here is about unifying the study of scripts using the active inference framework in all of these different dimensions the strong and the weak and the internal and the external and the social and the neurobehavioral all these different aspects of scripts how are we going to reframe it within the active inference framework and then what would that do what would that give us in terms of understanding or unique predictions or explanations in the abstract they also run through this notion that they're going to be leveraging the resources of active inference to offer up a formal model of scripts that's not just a flowchart or a mindmap or Pros but is actually a relationship amongst variables that specifies how some of these script uh variations like weak versus strong internal

versus external can be considered so we'll return to that in the figures especially and then in the road map we saw that classic formula with an introduction and then Theory X in this case script Theory and then active inference in section three and then in the following section A summation of theory x with active inference so that's the active inference account of scripts and this was an introductory work so there's probably many ways to follow down the rabbit hole for a specific subtype of script or other dimensions but um the main dimensions of variability in the scripts concept that were highlighted by the authors were this strong versus weak and internalist versus externalist so the strong scripts concept reflects event sequences that are extremely stereotyped or extremely reproducible or um kind of like that Grand Canyon in the landscape it's like you're being channeled along very strongly by a specific uh script whereas on the weak side we see things that are more probabilistic or potentially a preference or bias or more likely to go one way than another but it isn't exactly pre-specified and then on the y- axis we have this internalist versus externalist axis so the internalist perspective is going to look towards smaller things things that are inside of the person like the brain or the cells or computational things the cells are doing or something that the neurons are doing in terms of secretion or firing or electromagnetic rhythms whereas the externalist perspective as it just would sound like is thinking about things that are external to the focal agent so that would be the relationships in The Social Network as well as quote society and even broader things like the ecosystem cool so those were all things that we talked about in the previous weeks and now I want to raise some new questions about scripts relationships and active inference so the first question is how might scripts of different kinds whether the strong weak internal external or some other feature that people are wondering about how might they conceptualize of social relationships and in this case just for reference I'm thinking of this social network on the right there's two blue nodes let's just think of them as our Focus relationship so now we're moving beyond the individual as the focal system let's think about the edge and the relationship as the focal system and then seeing that edge in context of a broader Network so what might anybody think about um how would scripts and then I'll just throw up the second question too which is how could active inference add depths and tools or context to the social scripts concept any thoughts from any domain are good Stephen I might be curious like to hear about from a practice side how do how do we bring these ideas and then make them like tangible and experienced yeah I mean I I think this is this is actually where this scale question becomes kind of kind of it becomes a challenge right because as we try to M map things out um with social network so there's a lot of social network or

participatory inference map is one process which is quite useful for looking at and trying to understand how we um relate to each other's stories and experiences in our network but the question becomes is um how do we relate this um this kind of perspective we try to take on stuff with the scripts we use and then come back to well how much of those scripts are just manifestations of embodied inactive ecological behavior and we've kind of rified them so um I suppose um it becomes this we we we it's very much I think there's a kind of a um a situated um way that we we we kind of reenact or recapitulate this whole thing depending on the discussion so you know like if you're all sitting around a table like in that board meeting where people are putting together pictures of a puzzle they might be working with the scripts of how they should um assemble their perspectives and then you've got someone in a who's performing in a ritual and it might be much more about how they need to hold themselves as a person so I think there's something quite interesting that this whole scripts thing raises in terms of of the scale question and I suppose that's partly why you know Maxwell sort of ended up working a lot in here as well because I think it ties back to the question of what scale is the script happening at and that has a big impact when you try to then talk to people in a practice context because if you're working with a community or an organization or strategic strategy levels doing strategic foresight they they may be having their own kind of um scripts that shape their social practices in that kind of domain so I think this is this is kind of interesting what you just said there reminds me of uh two people like looking at a river and they're seeing different things they're seeing different scripts for The River there's no disagreement about what the river is doing but one person looks at it and says yep at the fork it goes left and the other person says it goes with the flow and today it's going left and so in that sense um they're seeing different scripts but they're not disagreeing about what actually is happening and so it's like there's multiple Concepts that are being projected on to the observables and then there's the story about those two people and how they reconcile their perspectives by the riverbank and so yes the scripts concept does seem to apply everywhere um but maybe to uh begin to Chisel away at what isn't a script I would say a chair could be a prop but I don't think a chair has a model where it is actually in agent doesn't have agency it's not an active entity in the script though it can be mobilized by actors in a way that a prop can be can become enlivened but it isn't with a generative model of that situation RJ yeah I was going to say yeah the chair the chair is an object that can be a part of a script but but it I don't think it can it's not a script itself and it can't act out any scripts right because it's not it's not an actor um uh unless maybe you know going out there it's like maybe if it were a mechanical chair of some kind

like a massage chair then it has been script that has been encoded by by some other actor um but yeah don't Sasha and then Stephen okay um that was a plot twist with the massage chair uh but perhaps uh the ability to um participate in a script does that does that imply that um you're able to respond differently in different situations because uh a chair will always be chairing um but an individual or a participate yeah participator in a script um they can change uh based on the context and the kind of script that they're participating in so maybe that's a key aspect of uh being a participator in a script Stephen yeah I think this chair is quite an interesting example and iose like in theater they often work with mask and The Mask I think they might be sort of bringing this back to the bodies like there's there's something about the mask and identity and taking off a mask and I haven't really thought about this before but I don't know say a throne compared to another chair could the throne almost be coated with a mask you know I mean it starts to have an identity and it's like you know so I mean a red nose is basically the smallest mask you can wear is basically become a clown but there all this work and we don't tend to do much with masks but if you go back to sort of indigenous cultures everyone was working with masks you know um so something interesting in is there something about the script which is about how we broadcast and receive through this part of the body thanks RJ uh yeah no no no so so um bouncing off what Stephen's saying um so one one of the things when I first read the paper I talked to you about this like immediately after reading the paper paper I think was um that uh if you look at uh there there's a common rituals between North American Indians Nordic cultures African like all over the place where where um what happens is there's there's it's either dances something along those lines where you use animal masks and act out the scripts that those animals use it's really important especially for you know like uh so I I brought up to you Dan at one point the bear uh in particular because bear is an animal that has a set of of of scripts that you can identify that it's using um and and if you don't have experience um you know knowing like reading their behavior then you're more than likely going to get hurt or put yourself in a really dangerous situation so like um park services in North America routinely make the point that almost all negative bear encounters happen because people don't know how to interact with the bear they run um they're they're not doing the right Behavior so even with like a like a black bear they play dead it's like don't play don't play dead with a black bear because they're going to investigate they're want to figure out like okay easy you know easy meal they're they're they're they're they're curious and they're going to go mess with you um so what what these rituals did was they gave young people who hadn't had the experience yet an opportunity to see those scripts in action

then before they may have ever interacted with the animal they may be asked to play out that script take the place of the bear wear the mask and and it gives you so acting out the script of the other person gives you a better sense of or or in this case the animal gives you a better sense of what it's like to be that animal so um in that case like acting out those scripts that other people may be using or you know other other act other actors would be using um you know actually is an advantage in the environment um on you brought up the throne though and it's like yeah it's it's a um well I I like I like the presentation of it as a mask for a chair uh but uh it's it's really important in denoting the person who sits in this chair should be acting a certain way right and and when they don't it's a joke right so I don't know I was just thinking about like like a few things there there there's a it's almost like there's a um a real advantage to understanding and then acting out the scripts of others nice R.J the hero with a thousand chaires as it were um I also didn't bring up wolves and I wasn't praised for that so exactly oh wait blue so just kind of thinking about like the chair itself is actually like supposed to act a certain way like you're supposed to do certain things with the chair right and and I mean I I'm always telling my kids like that's for sitting you've got the wrong end of you on the chair like so people get creative right like with the different roles that a chair can play it can be like a step stool or it can be you know some piece of gymnastic equipment and it's just it's just interesting to think about changing the script even for a chair let alone the person that's sitting in a chair it's it's like the setup for an improv scene and yeah the different kinds of chairs are going to lend themselves to different scenes maybe with a throne maybe they could enact a classical monarchy script or maybe it's a Monty Python it's a satire or with certain kinds of chairs or in certain context you can enact a shivalry script and you can pull the chair out for your colleague so there's all kinds of ways that um different props facilitate or enable different script to play out and you can always do a mime like you can sort of act out a uh digging or you can act out playing baseball but you do need the props our extended cognition to actually do some of these scripts that involve our extended Niche Sasha um I just wanted to touch on something Stephen said uh that um you said today we don't do much with masks um um I I guess I would disagree because it seems like uh a lot of us are doing a lot with masks and um just yeah kind of obvious to think that um that's a prop that is going to change our behavior in ways we might not even recognize and um yeah just important to think about um how this uh new prop is changing human interaction and scripts and um behavior in ways that we don't want or really expect uh a mask to change blue so Sasha didn't come right out and say it but I'm gonna like maybe point it at what you're saying Maybe not maybe I took it out of context

but um I I think that we do do a lot with masks like we are right now using the biggest mask ever like people are totally different in like their online Persona like some people will hide behind a computer screen and act in ways they would never ever behave in real life right so they have this like mask up they can create this new persona and and go forward in the world with that yep very true and with new props there's new scripts there's the one person's wearing a mask but the other one isn't script there's the is that person wearing an n95 script there's all these things that wouldn't have even made sense in 2019 so that's the sense making and the cultural sense making and the tension and the friction when there's differences on opinion or differences in behavior and it's about um not just estimates of state but actually uncertainty and estimates about other people's uncertainty that's this whole like recursive and embedded element of active inference and it's about navigating the uncertainty of a situation not just maximizing the state itself and that's something we'll return to for sure Stephen yeah I think this is interesting when you start to put the mask so this's I suppose I was talking about you know playing with a mask physically within this proximity of your face but yeah there's there's this kind of mask out there but it in some ways it's become abstracted and um like it's it's almost like like you can you put it on and put it off it's it's it's sort of almost part of the ecosystem that we've kind of created which is maybe a little bit unhealthy because we don't um there's actually one really interesting activity I did once there was a chat from Rwanda who does healing work and he would get you to put a mask on of your face and read a poem that was really personal and then you had to lower the mask but it was like you were lowering the whole of your sort of masking identity and it left you feeling this incredible rawness which is really interesting so I suppose um yeah there is this tendency now that and I think even maybe more since the 1980s if I look back at videos of people in the 80s on TV and that they're often a lot more nerv in front of the camera now people are used to putting up a some sort of smiley face type of thing and presenting I think there's um there there's something quite interesting in that that might relate to this paper yeah I've been thinking about that too like with the selfie new affordance of being able to at an arms length look back not just with a disposable camera and snap the pick and have the perspectival imagination of being at your arms end but actually seeing it in real real time and then video chatting or live streaming where you can see your own face during an interaction so it's like you're seeing the mask in real time and then that's sort of like um I think it's a Campbell quote or could be someone else but like uh if you think that you're rejecting the mask that Society gives you you're just wearing the mask of rejection everybody can see that it's not exactly the quote but basically it's

like you can't not play the mask game there has to be an epithelium there has to be a facade there has to be an interface and so we're talking about designing that interface because somebody who goes around saying No this is me this is me I have no mask this is just what I am it's like right but there still is an external layer that's being presented right and if someone says no I'm I'm All me there's no external level it's like wo big of true of course but then it it's not a helpful Viewpoint for the world because we do need to interface we need to send emotions reliably and receive emotions reliably and so um lot there let's continue with some of these fun questions so the third question is um how do the network relationships the red nodes influence potentially indirectly the focal edges in the network blue and then the fourth question would be like little bit more of a wild card what could be a fun or interesting social experiment related to active inference whether it's something that we do like a little experiment or just daily routine tweak or whether this is something that could be carried out with a larger scale laboratory online or in a social setting um so either of these questions people could raise their hand on I would uh start with that third one and uh highlight that that edge it's a relation but it's like an atom in the world of relational thinking the edge is like the atom and so now we're in this network of edges which is what makes it kind of a tapestry and a social Fabric and so if we want to understand the relationship between person A and B then we're always invoking all the other edges that stabilize them whether it's the who brings the food to these different people or how do they get their internet because they're not going to be able to connect if they're not going to get their internet and what are the Norms that they're getting the other 23 hours of the day and that's what shapes and in some ways can even Define the focal relationship is again the other 95% of the person's life or day that they're not in that relationship um during RJ all right so um uh just turning hand off there we go uh so on the last question I think I have um something I've been thinking about that may relate to the prior One on network relationships um so uh uh my dog Achilles uh he usually wakes me up in the morning uh for for like for last year or so uh we we go on a walk at at like 5:00 a.m. um and he is usually raring ready to go on that walk so I never really had to worry about an alarm clock he always you know he would always P me awake um and if I like there's no snooze button on a dog it is you know he he's there until I wake up um but recently after Thanksgiving um so I have Achilles and my parents have his brother Socrates and Socrates has been staying with us so now Socrates plays with him during the day wears him the hell out and uh ever since Thanksgiving um he doesn't wake up in the morning anymore he doesn't want to wake up he wants to sleep in so he had a script that was present in the morning that he

now no longer has and this has led to me oversleeping almost every day because I'm not I don't have any script for for interacting with an alarm clock in the morning anymore uh so uh what I think that uh you know going right back to the beginning of like what is not a script and and where's the transparency on like on on being able to look at look at what these things actually look like I think um our wakeup uh scripts is are are probably some of the most transparent because they're the most consistent um and they're the most um uh you know kind of there there's not a lot of uh outside interaction with those scripts it's it's it's uh very much like at the moment when you're waking up becoming conscious in the world so there's not a lot of Prior of them maybe your dreams Etc um so I think it would be kind of a fun experiment to to have people try to identify what it is their morning script actually is um identify where their first Network relationship is in that script right because maybe it's I know some people who wake up and immediately go on their phone um uh or you know like so so then um uh try to see if they can't pick out one part of that script that's not good for them that they don't like that's that's not healthy um uh and then like you know bringing back to my question which I wanted to resolve last last week was how do you identify the threshold events where you're able to change scripts I think it would be a fun experiment to try to figure out like like what are the best methods for changing your morning scripts which are definitely some of the most difficult to change good questions RJ thank you that um reminds me of somebody being asked what is your morning routine or what's your bedtime routine and I'm just projecting but a common response could be I don't have a morning routine it's like well surely you do have a probabilistic distribution of places you wake up and how often you brush your teeth even if it's 0% but there's some distribution of routines there that's the script it could be a weak script it could be a strong script but some people just brush off the question no no I don't have a mask I don't have a routine it's like a way of not engaging with that but once you recognize that it's about the interface and it can be a probabilistic interface just to say a morning routine it doesn't need to be regimented it doesn't need to be externalist it it doesn't need to be internalist we're moving beyond internalism and externalism as well as Beyond um um restrictive strong or loose weak we're taking all of those possibilities and now we're just thinking yes so what is your morning script set or maybe what what genre are your mornings um is it an emergency movie is it a comedy and then also this question about the dependencies and to take it back to this social network and the scripts it's like we often don't know all the dependencies until we do a perturbation so like RJ you didn't know about your lack of an alarm clock you know stimuli or script until there was another script that changed in a non-human animal and then

that had a cascading failure in this network of dependent scripts so it revealed new relationships and dependencies because of a manipulation and so that's very related to active inference on a few levels just to provide one is we're always interested not just in coming to Clarity on world States but actually coming to Clarity on which interventions or which experiments or perturbations would maximally inform us while also remaining functional about those dependencies so then you could be um you even when the dog was still waking up early you could have been experimenting with earlier or later wakeups and maybe by uh bumping things by half an hour this way or another you would have discovered oh wait without the dog I don't have a way to wake up or if the dog is tired then there's this other relationship and nobody would connect the dog being tired to waking up early for most people especially if they don't have a dog but in this case it could have been explored or actively inferred with um rightful experimentation Sasha um yeah that's really kind of funny about the morning routine because um yeah we we of course all have one but maybe it's uh not not very glamorous or uh effective and it just reminds me of um these sort of uh glamorized morning routines of celebrities where like they spend um you know all this time putting stuff on their face essentially and uh it kind of signals like I have the kind of morning routine that I'm proud to tell other people about or should be a you know a sensational YouTube video that other people Envy um and so kind of changing yeah what a morning routine is or should be and then also a part of that whole like um kind of uh thinking about your morning routine is that it's very solitary when in fact of course we wake up for other people or other creatures and that um we kind of get sucked into someone else's script even before we know that we're part of it um like people who wake up to take care of their kids like maybe they don't want to be waking up but you know um and then how to use active inference in a way that um helps people modify their um morning script um in a way that I I guess it just kind of reminds me of like biohacking and and um being very like uh focused on every metric that you have so do you have to have all those metrics in order then to change your morning script or like to optimize that or is there a way where active inference can help us get around that um kind of uh self-measurement Obsession that biohacking can fear off into um when it can be more focused on just uh yeah the the performance and the human script that we're participating in good question there Sasha and it reminds me of just the imperatives in active inference being to reduce uncertainty and to design optimal experimentation and so instead of um in this reward framework thinking we're going to have the most rewarding morning thing it's going to set you up for the most productive day or it's going to feel the best it's all about optimization most

most most and here we're actually saying yeah maybe we'll get there that'd be great but just where we are how could we reduce our uncertainty like maybe every morning you wake up and you're not sure whether you're going to take mechanism A or B so how could we start by reducing uncertainty and decreasing anxiety about the coming day increasing Clarity and then asking not what is the optimal state but what's the optimal experiment and policy that's within our scope of affordances not like waking up and instantly going to the strange attractor of thinking that something could be a different way um we can focus on our policies and then this Beyond internalism and externalism it's just so critical because it's uh sort of the false dialectic of like worrying about oneself only and then disregarding systemic issues versus getting extremely obsessed with systemic issues and then losing the sight over the affordances that we do have and in the middle um is the compromised language but we're going to not just be in the middle we're going to be all over the map and we're going to be able to dive into the internalist thinking when we need it we're going to be diving into the externalist thinking when we need it but we're always going to be able to return to the fullness of the niche and what's possible and getting other people's perspectives on the issue so that's I think a a a market difference in active inference um let's uh go on to the next slide so this one's just kind of a little fun single question I was just wondering what does a script look like and so the movie script this is the Lethal Weapon 4 autographed Edition script um one of a kind uh that's a movie script so the words are syntactically encoded and maybe the semantics are suggested or it gives stage directions like a dramatic script and then in the middle is this like peer pressure script like what's the script that keeps people driving on the right side of the road in America and on the left side of the road in the UK that is something that's enforced by Semiotic cues like Signs and um counterfactuals like if I were to drive on the other side of the road I would be in the uh felony script or something like that but how does pure pressure play out and how do we visualize it and then just a big unknown is what do these active inference scripts look like like how would we communicate the scripts and so instead of um using a movie dialogue to convey some of these probabilistic things or the peer pressure um narrative to talk about some of these things is there another visualization or representation just I'm not sure if anyone has thoughts yeah RJ or and then anyone else go ahead uh well I think um uh I I don't know if um I'm pretty sure I'm pretty sure the graphic was was in was in the um was in the slides uh last time was like uh kind of you know just a you know 2D representation of the axes of like strong script to weak script internalist versus externalist and um you know the the paper itself like like tried to show that it's like look these these are just um you know these

don't have to be Divergent views they're just different perspective yeah that's it yep um and I feel like uh you know the the actor script is just a a it's a Continuum it's very strong externalist script you're either performing The Script exactly or it's not it's not it's not being performed it's not happening um then you have like yeah I brought up Always Sunny last time it's like it's a show that allows for a lot more improvisation so the scripts they use while strong and externalist aren't as strong as say Lethal Weapon right um but then peer pressure is something that that's that's far more internalist your your your interpretation of what you think the script is and maybe talking with people about what you think the script is and trying to work that out and it's a weak script where it's like you can even see there that like in in the picture that it's like not everybody is acting out the same exact script and like you know um like not the same exact movements whereas you look at something like um you know uh opposite end of of the the um the the political representation maybe represented is like you know um uh or or I'm trying to think like historically uh like like um uh you know um during during the Third Reich it's like these were mixes of um of internalist scripts of what you think you're supposed to be acting out but then there's also these strong scripts involved where it's like you are supposed to make exactly these motions Etc Great points there and yes the pure pressure I put it up there partially because it so directly calls us to move Beyond internalism and externalism I mean what could be more external than other people yet at the same time what could be more internal psychologically or neurophysiologically than your interpretation of peer pressure and then on the strong versus weak just like you brought up um what could be weaker than an invisible and lightly guiding force in some cases like hey peer pressure is like you should wear uh shoes that have tied laces but no one's going to like freak out but there's other peer pressure things that people will freak out and that's a strong script if 100% of the time that something happens socially enacted another thing happens it's quite strong because 100% of the time from State A it moves to State B mediated by an external Force so that's strong and external but there's other cases where it's weak and internal at least in how it appears and again the goal of active inference isn't to be like yep we judge strong versus weak and it's strong it's rather to see this Continuum as um an area of theoretical dispute yes but now just a parameter in our way of thinking about the world so that there can be subcomponents or different people in the same crowd who are experiencing very different realities or having very different um sort of generative statistics for their decision-making based upon some of these features we're discussing but yeah very interesting any other thoughts on the visualization otherwise there's um there's more we'll talk to because um it'll be

good to have time for these two figures is that good all right let's go to these um two figures in the last half hour and just start out with just what are they showing because especially for those on the social sciences side it might be like whoo interesting conversation about the scripts and the chair and the being and the becoming but what are these variables and circles and lines so let's think about making that bridge between the qualitative things and the experiential and the social that we've been discussing with a more mathematical way that we're going to be returning to in all these different conversations so and as always people can just raise their hand but we're going to do maybe 10 to 15 minutes on each of these two figures so figure one in this paper is a simple generative model for policy selection and in fact INF policy is always P_i for policy so this is what is being estimated the schematic depicts a generative model for policy selection it represents probabilistic beliefs about and then all of this is colorcoded so the first thing is a A is a probabilistic belief about how observations are related to the states that cause them so here's observations this is like red Photon on my retina and this is state is like I'm wearing a red shirt okay so then there's this mapping in this case it's pretty simple where if there's a red shirt and the lights are on then I'm going to get red photons on my retina but one could imagine that there's State estimates that are social like this person is unhappy with me and there's some probability mapping onto different observations of body language or of um words for example so that's the A matrix it links the state estimates of the world to observations and A happens instantaneously in time now B is where we start taking lateral moves and this diagram goes from the left initial conditions to the right through the time steps 1 2 3 this is discrete time and B moving us from left to the right is mapping States onto each other so if we were talking about the red shirt well if you're wearing a red shirt at one moment basically 100% of the time you're wearing a red shirt at the next moment so that's one simple format or it could be again social like given my estimates of this person what do I estimate the transition Matrix is so knowing this person they get unhappy really fast but it's really easy to bring them back to being happy or they're always happy but once they get unhappy they're going to stay in that state for a while so you can imagine like a matrix where the on and the off diagonals are different values and that captures whether once somebody is in a certain state is it an attractor state or a dwell state is it a state that they evacuate very rapidly like balancing on one foot so that's what's captured in this B Matrix is how things change through time um D is the initial conditions so those are states prior to sampling the world so that's like when we start having a conversation I am estimating your initial uh condition for that conversation um it's not that I know nothing about you I'm

bringing in my estimate of before the conversation before I started sampling in this EPO where were you at and then in um the top we have G which includes C which is the preferences in the outcome matrices and so G that's definitely um uh unpacking equations for another day for 11 as well as for other discussions but let's just clarify this upper half G Bears on the selection of policy that's why in active inference we're so obsessed with free energy and how it relates to control theory cybernetics policy selection because all of the coolness of free energy is that it's a different way of thinking about how policy selection occurs so that's why G has an arrow pointing down to Π now Π is influencing only B's so this is what's pretty interesting is what is being estimated by the agent is like this whole graph okay but the connections within the model that are relevant are not from policy to state so it's not saying how will States be different as a function of me with this investing strategy or of me with this social strategy or me with this navigation strategy how will the states be different actually it's policy to State transition matrices so this is like saying if I undertake a policy where I exercise every day then I believe that the the state transition Matrix from being unhealthy to being healthy will be modified not that I will move from unhealthy to healthy that's the state estimator that'd be policy to State and it does result in the same conclusion functionally but it's actually very very different from a mathematical and from a programmatic standpoint to say that the policy estimate Bears on how States transition into each other not just at the first level of what the states will be so again I can do policy one two 3 four five and instead of saying well if I do policy one then I would estimate that this state is most likely and if I do policy 2 this state is most likely that's the old way of thinking the B Matrix way of thinking is under policy one the world is apt to change in this way under policy 2 the world is apt to change in this way and then that is related back to affordances and preferences in the C Matrix as well as other components of G that include for like the Curiosity and the maintenance of um rewarding States and all these other features so what this graph is showing again just to sort of close out this little section is we're unpacking this idea of policy selection so just like if we were picking out how to drive from point A to point B we're unpacking this policy selection with an eye towards social interactions so where the state estimates are like whether the person is happy or unhappy with me what are the observations I'm going to see and how do those get mapped to each other what is their state through time or my state through time and then how does that change what were the initial conditions that set up the first observation we got and then as a function of a lot of cool stuff the sort of bulk of the iceberg in G how can policy be selected to shape the way that states transition to each other in a way that's most aligned with

how I want the world to be and then agents that succeed and that succeed and those that don't don't so I hope that that sort of conveyed some of the attributes of this graph any thoughts or questions or what what's another context where this graph could be sort of lifted up and placed down upon yeah so while people are thinking and raising their hand um again I would just try to give one more Bridge to the onboarding to active inference side because for those who are familiar with the formalism this is not even the full formalism they've probably already seen more expanded versions but for those for whom this type of diagram is relatively novel in a social science setting just always keep in mind that no matter how many letters how many edges superscript subscript Greek letters you know cyc letters whatever there is we're trying to just be specific about what we're talking about and find a generalization or a structure that allows us to map structurally analogies from different social settings so this could be um so many different systems will fit under this type of model as well as and importantly that strong versus weak internal versus external idea that we were talking about earlier so we can have the initial conditions be me going outside with no shoes and here somebody else estimating um things about me or me estimating things about other people or this could be um the initial conditions of Hamlet in a given production or this could be the initial conditions of 2020 and so how in all of those different areas that qualitatively we were talking about how scripts could be used how are we going to find a sort of uh skeleton scaffold that all these other situations will be able to kind of hang on in a way that respects the specifics about their situation um as well as might even provide us with new knowledge so for some of you that might be hearing it twice the same way in a row for some of you it might be zero and maybe for others it's in the middle any thoughts though before we go to the next figure which is probably a bit richer all right let's look at the next figure so anyways this was the skeleton generative model so hopefully we just familiarized ourself with the idea that things about social situations could be abstracted and represented with uh vectors and matrices and policy now we're going to take that to and here's in the paper where they write about how that could map now let's look at a little bit more expanded of an example okay so now we're looking at not just the agent but also the agent and the world or what are called here the internal realizer and the external realizer cuz in figure one one we were just looking at basically the internalist perspective we were just looking at you know you could imagine that a brain we're doing that or a computer we're doing that but there's nothing about the world there's no there's an estimate of the world state but there isn't an actual World state in this model so this is purely internal to the agent now let's look at figure two figure two is where we're going to start to bring in the world and we're

going to bring it in at a very specific position and that is generating the observations that's really crucial the world doesn't intervene by jumping into the internal model now it might in reality this is the instrumentalism of active inferences we're going to model it as if the only interface that the world could have is with observations and so in this figure one the only inputs to the model were basically the initial conditions and then the observables and now we're picking right up here at the bottom of the figure and we're going to graph the world in so here's those observations and those observations are being generated by World states which are 11 so here reading from the um caption on the side of the external world there's only one process in play which is the likelihood of observation States the world 11 and the transitions between these states in nine that's down here okay so this from the point of view of the agent is the generative process that's creating the observations so like there's a shirt that's generating the red photons that is hitting my retina for example but here's what's really interesting is this model of hidden State mapping observation transition Matrix change in state emittance Matrix observation again that is what we just saw in figure one flipped 180° State estimate and transition Matrix and then leading to observation that is this deep and fundamental symmetry that has been explored by axle constant yell brunberg and others it's this idea that we're going to do this mathematical phrasing where like we know which side we're on we know which side is the internal side and which side is enclosed by the epithelium but mathematically look at 75857 and similarly 11 10 9 10 11 here you just have have observation in the middle it's on the interface that's like the retina being hit by the photon and then there's a mapping on the internal side from the agent through some type of likelihood Matrix to the observations so the agent is like reading it's trying to say what is the hidden State five given observation so what is the best seven to estimate five given the observation and then what is eight what's the transition Matrix from 5 to five and then similarly for the next time step and then on the other side it's the exact same except the key difference is that in the case of the internal agent we have six and so in six we have policy here it's a flirting policy but in the world there are things that change state to state in the world but we don't have policy control we don't have agency over those States and so then in order to select policy we have that g and that's kind of where we see the similarity to figure one so um it's just a very interesting formulation I'm sure each person's going to see their own thing in it and let's you know have some conversation or questions about uh what's unclear here but we're setting up the world and the agent theistic description of the generative model of the niche and of the agent we're setting them up in this intertwined and uh inextricable but also deeply symmetrical iCal way where if it was an agent

who's like an organism and the niche is the forest so the forest doesn't enact any strategy it has a certain layout of acorns but that's not a strategy that's like a chair for the squirrel and then the squirrel is the active inferring agent that's going to be doing things in this Forest so we can capture that here but also if it was squirrel versus squirrel combat and each squirrel was going to be having a generative model of the other squirrel having a generative model of it these are higher order theory of Mind recursive theory of mind it would be embedded in recursive because it'd be like about other active inference agents in the world and it could be deep it could be temporally deep it could include counterfactuals it could include affect or veilance that's what we'll be talking about in 11 but it's set up in this way that's very parsimonious because in the limit it allows for a true symmetry across the interface across the blanket in the Niche where the niche can be learning and making a model of the agent and vice versa so in the extreme it allows for that type of richness of agent agent Dynamics or this could be reduced to two inanimate objects so if there's two rocks that are just you know two bumps on a log and they're not doing free energy minimization or recursive estimation of anything they're just doing what they're doing with a certain State transition Matrix that can be captured within a reduced version of this framework so yeah not trying to rant on it but I just think that's kind of an interesting piece of it and we often come back to this idea of Niche construction and it's like wait isn't control theory about what's the best strategy where is Niche construction coming into play internalism externalism it's like we're moving beyond internalism and externalism because we're saying you always need both you're always going to need both and there's going to be many types of interfaces one important interface Bas is seven and the observations which is like the sensory States the sensory cap capacities of the organism the umelt and the actual observations themselves that's like the handshake or the API call that actually crosses the interface those are critical but almost by definition because if there's something that doesn't cross the interface on the inbound side like a sense data that doesn't hit a neuron or on the outbound side like an action that doesn't act if it doesn't cross the interface and make the connection it's almost like it didn't happen and so we Define the interface in terms of the connections that do happen and then we can always be sure to be talking about relevant features of the world and always in a way that moves Beyond Simple conceptions of internalist versus externalist will have the ability to find situations where the internal features given a certain blanket of analysis dominate the Dynamics of the system or where given the inclusion of certain parameters or in a certain time scale of analysis the external features dominate it but no one can say that's really a win for externalism or a win for internalism it's

just a description of a situation so um I think it's an example about how we move Beyond some of these theoretical um and literature grounded abstractions with a more abstract and general framework that actually lets us be more specific in the projects so we kind of escape this uncanny valley of whose team is better internalism or externalism we're going to escape that Valley by taking one step towards abstraction by like whoa what if we thought about the agent in the world as acting across an interface in the most General possible way but it turns out that by Framing it as an interface we're going to be able to be very very specific about the narratives of radicalization or the kinds of cues that are exchanged in a flirting relationship whereas if you're in the debate of internalism externalism you could have whole academic conferences on narratives of radicalization or behaviors of flirting and potentially the people would leave that meeting still feeling like their side had won so there wouldn't be a true conceptual synthesis so again a little bit another Spiral away another tangent but that's why these kinds of formalizations and specifically graphical and computable ways of thinking about them can bring Clarity to some of these long-standing qualitative questions and also necessitate the collaboration of basically quantitative and qualitative expertises RJ um so so um I've seen like not exactly this this graphic but but um uh throughout each of the papers especially the ones first's involved with there's always some iteration of it um and some iteration of explanation of active inference and one thing um that I've seen alluded to in the words but not in the graphic is um this updating of the observation itself um and maybe this is being represented here and I'm not seeing it um but but what what I found is like um and and I'll get to why this is uh fairly important to me in a second but but um it seems like well in any process of sense making we have this induction and at the state of induction um you're actually like or you're getting to trying to parse meaning you're you're you're trying to like check whether or not what you've observed is even the correct observation so it's the double take um and and this is one of the reasons in intelligence analysis that like collection is separated from from from your your all other elements of the production cycle because that what they're trying to do is they're trying to make sure that there's not this like constant seduction of going back and checking again checking again checking again they're trying to make it so there's a pipeline so when something comes in it's going up the pipe it's it's not your job to look any further into whether or not what you saw was meant something or whether or not it was correct or it's just report what you saw push it through and now it's not your problem anymore it's up it's up somebody else's problem right um so like now at at the at the the scale of the individual though um we're constantly double- cheing ourselves whether or not we

we saw we thought we saw so like if I'm if I'm reading a paper and I find um you know um something cited and and and I kind of know where that's coming from so this happened when we we um few people on this call we were working together on on on a paper for active inference and behavioral engineering and I was looking into a spirit decor and while I was doing that there was a big problem because there were so many different interpretations of what a spirit Decor meant so so what ended up happening is every time I saw the words I'm checking myself and going wait is am I actually observing what I think I'm observing I'm checking against priors and I'm updating my observation um so what what I'm kind of wondering is like um one like is this represented in this graphic and I'm not seeing it uh and then two um I know that over time we're starting to see okay like actual computational models um using active inference are those models like if they were applied in an agent-based model or applied in machine learning are those models trying to act that process out of like checking back on the observation and making sure that it's actually observing what it thinks it's observing um rather than just taking data in does that make sense yep thanks for the question RJ so it is lightly indicated in this figure as with the other figure there's a movement from left to right okay so the nodes are numbered by the kind of things that they are and so it's lightly indicated where the direction of time is going with these dotted arrows so like for example the first 10 on the bottom left that's like a hidden state of the world right and through an 11 it emits an observation at time point one then there's a transition Matrix nine and so if you did um hourly checkpoints the nine Matrix that you calculate is going to be the hourly transition Matrix if you do one minute checkpoints like a discrete time with an interval of one nine is going to be a different Matrix so it's kind of like how the the candle chart of a stock or a crypto looks different depending on what time length the candle is so it's not the same Matrix for all time intervals it's always model interval dependent but nine moves us from the left 10 to the right 10 and then at that 10 again it's the type 10 not like node number 10 it emits through the same or a different 11 another observation now the early models all the 11s were fixed so like the hidden State could go from daytime to night time but like the 11 was fixed and then you could imagine that a more advanced version of the model would be like yeah actually the hidden State through time it changes in how it emits 11 okay now flip the interface seven is perceiving the observation the left seven and then it's doing estimation of the hidden State s so that or five it just looks like an S and it is s in the models so it's kind of like at each time point we have this like 10 hidden state of the world emits through 11 in observation that's on the interface that's that holographic boundary that Chris Fields is going to help us understand then

on the organism side the internalist side seven does a state estimate five then we move through time the organism updates its estimate through time with eight and the world updates its actual state with nine it's hidden State and then there's another time Point another stack of that and so in the example that you gave with like going back and double-checking information it's like you get an observation at one moment and then whether or not it's concordant with um something at least you're instantaneously sure of what it is but then here's another dimension that the computational models are going into which is at each time point it calculates an integral through time of policy so that's the part where we're going to talk about it in 11 and a few other ones but the simplest model would be like again red shirt or you know black shirt and then it's like the photon is the observation and the state estimates pretty straightforward it's like a 2 x two Matrix there's an unchanging characters in The Matrix because the red shirt always emits red and there's no ambiguity okay now we can start to relax those assumptions and we can think about okay maybe there's more ambiguity in the signal maybe there's a non-stationary 11 Matrix maybe we can learn seven we can learn better about how observations are related to World States maybe we can learn eight better which is sort of like uh trying to reflect nine we can learn about how states in the world transition better and we can learn better policies too better sixes but that what we've just been focused on is really just the loop with the 10 n 10 n in the world 58 58 in the agent and then this interface that has a symmetry of the sevens and the 11s around the observation itself so yes they're the kinds of things that people are starting to roll out but I think that especially if you watch 11 coming up you'll get a sense at how rudimentary these rollouts are really it's at like the uh IBM chess 1995 status where like many many branches of counterfactuals are being processed and then evaluated and more modern chess computers don't use this Branch rollout evaluation strategy in the same way so we're still in a very early stage that's basically um defining what that roll out would be through time and then there will be other ways of heris approximating very good rollouts Sasha yeah nice thanks for walking through that on the diagram uh with all the numbers um yeah uh just some things that RJ reminded me of um is in thinking about uh data collection uh it would be great if we could separate um the data collection from the data interpretation and acting on it and um you know informalized uh kind of scientific approaches we can do that like um I'm just thinking about my own work when I do uh microscopy um and I'm looking at some cells under a microscope and um I might have like an emotional reaction to how they look which is like they look really good and I want to analyze them or something went terribly wrong and they look like um not even cells at this point so um I think part of uh going through this

process is being able to separate the individual steps so I have to actively prevent myself from uh yeah uh kind of throwing in the towel preemptively when I don't uh that when the data doesn't meet my expectations until I go to the next step of like randomizing and analyzing it um such that I can remove my expectations um and just try to collect what is there um and I think that uh yeah that that's something that's nearly impossible to do in The Human Experience that we can't really separate um the data collection from the data interpretation and um yeah it just moves too fast and then in the um realm of like uncertainty and um yeah what does this word mean in different contexts as R.J was saying that is the kind of thing that um in teaching like uh we try to think you know if you don't know every single word in what we're learning today um it's going to be really hard to build on any Next Level Concepts so and students will often say like well I don't really know this word so I don't know but the first thing to do is to like just look up what the word means so that you can move past that uncertainty and move on to move on to the next layers of things um and so that really to me makes me think of like social anxiety and uh people not being able to interpret social cues appropriately and that creating uncertainty which is just like it kind of snowballs once you're uncertain about one aspect of an interaction or a script um you kind of are prevented from participating in the next levels or carrying out the script and so that perhaps is something interesting to discuss um in next week's interact uh discussions cool thanks I think that question especially Sasha really makes me think about like script design and we're talking about the analysis and the descriptive ways of talking about systems but also about designing the scene and the stage um I think Steven dropped off but is somebody with uh drama experience performance experience you know social performance performing our roles a lot of these terms are starting to blend with the mask and previously that would be seen as dramatic or ritualized now it's seen as a matter of public safety so many of these things are actually coming together in a way where we can um certainly within our own life if not at a higher level ask about what kind of scripts we want to design for moving forward especially about the uncertainty from all these different issues so um here's just a couple of questions I wanted to throw up we only have you know 3 minutes so really just for the listeners live or in replay to add comments or start a discussion in whatever platform we're using just questions like thinking about how this could be a unifying topic how could this be a valuable model what unique insights are we looking for how could we rethink our experiences what would be explanations or predictions and what would be next steps for the research and um thanks for participating I'm going to just pause it for um on this final slide as we hear our last thoughts so

everyone prepare a little last thought if you'd like and to listeners thanks for listening we will have follow-up forms for the live participants it's in the calendar invite and anyone is welcome to provide us with feedback and stay in touch but for the last 3 minutes let's just hear from our panelists anybody who would like to what is a last thought what's something you're still wondering about or now that we're done with 10 where does that leave us what are some last thoughts R.J and then blue uh so we I think we we did find some conclusion on um what can to use a script um but uh I still don't feel like we have a good conclusion on what a script isn't what is that line so um I'm going to be thinking about that the next week and I hope we we touch on that next discussion awesome blue so I'm just wondering left really wondering where can we use this to engineer social behavior or to change scripts that are perhaps unnecessary or unwanted in society like radicalization scripts or or things like that so that's where I'm left you know how do we then go out and apply what we've learned cool um either of the Sasha or Ivon um yeah I similar to Blue am left thinking about what is the application of active inference in scripts and um I guess in instead of feeling uh uh unsatisfied I I feel excited about it that's my perspective um that it seems that maybe at the end of the last conversation I felt uh ambivalent or confused about scripts and now I'm a bit more uh directed in their their use or their possibilities so I'm I'm really happy to kind of have uh talked through this and uh find a way out that makes me feel excited about scripts that hold space specifically for things like freedom of expression or speech or improvisation it's like um a protection within the script this is the part of the song where there's a drum solo so for this part you get to play in time and then for this part it gets to be wacky and so in that way we can um and like blue is saying as well with the design we could be able to converge on scripts that protect certain freedoms as intrinsic to the structure of the script and being able to have a discussion about that that is um nuanced like we hopefully are representing and learning how to represent here that could be a really productive discussion instead of just about rights versus um all these other things that can get tangled down in so everybody thanks so much for participating and for the great discussion as always this is it for 10 we will have 11 in the final two weeks of December 2020 the 11th Hour the 11th act imp stream and then we will have 12 the clock will strike midnight on actm 2020 and we will be heading into a very exciting 2021 so thanks again for listening and for participating and see you later